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Journal of Experimental Marine Biology and Ecology 317 (2005) 53–68

Journal of EXPERIMENTAL MARINE BIOLOGY AND ECOLOGY

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Spatial, temporal and interactive variability of infauna in Swedish coastal sediments

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Received 26 March 2004; received in revised form 16 September 2004; accepted 3 November 2004

Abstract

The relative sizes and importance of spatial, temporal and interactive variability of soft-sediment assemblages in Swedish coastal waters were estimated using a hierarchical sampling programme. Spatial scales ranged between 10^{0-1} and 10^3 m and samples were collected at three times within 1 year. Variability in diversity of assemblages, as well as abundances of the 15 most common taxa, were estimated and tested using ANOVA. Nonparametric multivariate analyses of variance (NP-MANOVA) were used to test and estimate the variability of whole assemblages. Univariate analyses revealed that all variables showed significant spatial and temporal patterns but that these differed among individual taxa. A general finding was that there was much interactive variability between times of sampling and the two spatial scales tested. This means that abundance and diversity of infauna changed differently among bays located a few kilometers apart and among locations within bays located hundreds of meters apart. Thus, spatial patterns of assemblages changed among times of sampling. Conclusions were similar for univariate as well as multivariate analyses, although multivariate patterns appeared slightly more persistent. Furthermore, estimation of variance components showed a substantial variability among cores within locations. The results provide the first coherent analysis of interactive scale-dependent variability in these geographic areas and one of few in soft sediments. Results are discussed in relation to previously observed patterns of spatio-temporal variability in soft sediments and in a context of potential processes responsible for creating patterns at particular spatial scales. Results were also used for cost–benefit analyses in order to compare precision of optimal solutions to what was observed and to optimise future sampling of these assemblages.
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Keywords: Analysis of variance; Benthic fauna; Scale; Soft sediments; Variability

1. Introduction

In spite of representing a small proportion of all marine areas, the coastal zone is of fundamental importance to the development and sustainability of human populations and societies (e.g., Suchanek,

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doi:10.1016/j.jembe.2004.11.014

1994; Gray, 1997). In a broad sense, the diversity of valuable goods and services provided by coastal areas is related to its large diversity of environments, habitats, and species. Shallow coastal areas are often dominated by a diversity of sedimentary habitats, which harbour a large diversity of animals living in and on the substratum. Apart from its inherent value as a resource of biotic diversity, macrofauna contributes in a multitude of ways to ecological functions, which are considered useful from a human perspective. These functions include enhancement of geochemical activity (e.g., Aller, 1982; Kristensen et al., 1992; Gattuso et al., 1998; Rosenberg, 2001; Lenihan and Micheli, 2001; Solan et al., 2003) and basis for production of fish (e.g., Pihl, 1982; Heck et al., 2003). Understanding the patterns and processes associated with this diversity is therefore of great importance for coastal zone management and conservation (e.g., Lubchenco et al., 1991; Snelgrove, 1998).

The study of spatial and temporal variability has been the focus for much recent research on patterns of benthic assemblages. In ecology in general (e.g., Denman and Gargett, 1983; Kolasa and Pickett, 1991; Lubchenco et al., 1991; Levin, 1992; Schneider, 1994), as well as in marine benthic ecology in particular, there has also been a strong emphasis on research into the relationship between variability and spatial scale (e.g., McArdle and Blackwell, 1989; Thrush et al., 1989; Thrush, 1991; Morrisey et al., 1992a; Lindegarth et al., 1995; Underwood and Chapman, 1996; Hewitt et al., 1997; Kendall and Widdicombe, 1999; Bergström et al., 2002; Hewitt et al., 2002; Ysebaert and Herman, 2002). This is because information about the magnitude of variability at multiple spatial scales has potentially profound implications for our views on the relative importance of ecological processes structuring benthic populations and assemblages (e.g., Butman, 1987; Archambault and Bourget, 1996; Legendre et al., 1997; Underwood et al., 2000), the design of tools and guidelines for monitoring (e.g., Underwood, 1992; Inglis and Lincoln-Smith, 1994), as well as management and conservation (e.g., Rykiel, 1998; Garcia-Charton and Perez-Ruzafa, 1999).

The conclusion that emerges from most studies of assemblages of macrofauna in soft sediments is the prevalence of scale-dependent spatial variability in abundance and composition, despite apparent general environmental similarity among areas, and that the relative importance of different scales of variability differs among species. This clearly indicates that ecological assemblages are affected by a complex set of processes, which vary in intensity among scales. One insight from the current focus on spatial scales is that speculations about the importance of a certain process in terms of its influence on local density and composition of an assemblage are only meaningful with respect to certain specific spatial scales (e.g., Butman, 1987; Archambault and Bourget, 1996; Constable, 1999; Thrush, 1999; Zacharias and Roff, 2001). Obviously, evidence of the importance of large-scale processes such as ocean currents and salinity gradients cannot explain small-scale patterns, while patchiness of sediments or interactions between larval behaviour, local topography, and hydrodynamic processes whose intensity vary at small spatial scales are not likely to powerfully explain large-scale patterns of fauna. In this context, observational studies of spatial variability of populations at multiple scales may provide useful perspectives for assessing the overall importance of a certain process (e.g., Constable, 1999).

The fact that there may be large temporal variability in abundance and composition of assemblages in soft sediments is well known. In boreal areas, there has been a particular emphasis on seasonal variability (e.g., Beukema, 1982; Möller, 1986). Studies explicitly exploring the relationship between variability and temporal scale are, however, scarce. Some of the available evidence suggest that there exists substantial unpredictable variability at surprisingly small temporal scales (i.e., days, weeks, and months), which, to some extent, may obscure assumed predictable fluctuations among seasons (e.g., Morrisey et al., 1992b; Olabarria and Chapman, 2001; Lindegarth and Underwood, 2002). Another important aspect of temporal variability is that it may interact with components of spatial variability (i.e., the change in abundance and composition from one time to another may differ among places). In terms of deepening the understanding of structure and dynamics, the presence of such interactive variability is of fundamental importance for conclusions about the general importance

of any explanatory model (e.g., Underwood, 1997) and stability of benthic assemblages (e.g., Turner et al., 1995; Edgar and Barrett, 2002; Ysebaert and Herman, 2002).

Because of differences among studies in the resolution and range of spatial and temporal scales, analytical procedures and structure of benthic assemblages, and synthetic statements about the importance of interactive variability are difficult to make. Nevertheless, it is clear that the relative emphasis on stability vs. variability of assemblages differs among studies. For example, at six 9000 m^2 sites in New Zealand, Turner et al. (1995) found that there were marked temporal fluctuations for some populations, but that the communities were generally persistent for a period of 5 years in terms of overall community structure. From yearly measurements during 6 years, Ysebaert and Herman (2002), on the other hand, found that components associated with interactive variability were statistically significant and ecologically important in Dutch intertidal sediments. These studies indicated that yearly changes frequently differed among stations 10^2 m apart, but more seldom among regions and transects 10^4 and 10^3 m apart, respectively. The results also suggested that spatial variability was relatively more important than overall variability among years.

The aim of this study was therefore to estimate the relative importance and test the statistical significance of spatial (at scales ranging from 10^{0-1}–10^3 m), temporal (within a year), and interactive variability of soft-sediment fauna in a Swedish coastal area. This was done by analyses of abundance of individual taxa and various measures of diversity, as well as analyses of overall structure of assemblages using multivariate techniques. Previous studies in the area have primarily focussed either on spatial or temporal patterns of certain conspicuous taxa (e.g., Möller and Rosenberg, 1983; Lindegarth et al., 1995). Therefore, this study was designed specifically to assess and quantify the relative importance of interactive variability at several spatial scales. Such results provide a valuable background for further tests of hypotheses about the generality of ecological patterns among scales, potential structuring processes, and predictive modelling.

2. Materials and methods

2.1. Study area

The investigated region is located in the northern part of the Swedish west coast. This area is influenced by currents transporting water from the North Sea and the Baltic. The maximum tidal amplitude is only about 0.3 m, but due to air pressure and winds, the water level fluctuates with an amplitude of approximately 2 m. All fauna samples were collected from unvegetated soft sediments at about 1 m below mean water level. The five bays sampled are of about equal size ($\approx 10^4$ m^2) and, on average, subjected to a fairly similar degree of exposure. The north–south extent of the studied region is about 4000 m (Fig. 1).

2.2. Sampling methods

The choice of core size and sieve mesh was based on a pilot study conducted in the area. Two sizes of sampling corers (diameters 5 and 10 cm, respectively) and two sieves (0.5 and 1.0 mm) were compared. The corer was pushed 10 cm into the sediment and carefully dug out. The sediment within the corer was immediately transferred to 1.5 l plastic containers and transported to the laboratory at the end of the day. The number of taxa collected in the large corer and the fine mesh was two to three times larger than with a coarse mesh. Comparisons between the larger and the smaller corers showed that the number of taxa was twice as large with the large corer compared to the small. Coefficients of variation (i.e., standard deviation/mean) among cores meters apart were 0.6, 0.6, and 1.3 for the total number of animals, polychaetes, and bivalves, respectively. In subsequent sampling, we decided to use the large corer in combination with the fine mesh in order to minimize the coefficients of variation and maximise the number of taxa collected. Thus, the animals were sampled with a 10 cm diameter corer, extracted by sieving through a 0.5 mm mesh in the laboratory and preserved in 10% buffered formalin. Animals were sorted and identified with the aid of a stereomicroscope. Animals were identified to intermediate taxonomic levels, usually families. This is generally sufficient for identification of natural and manmade spatial and temporal variability of assemblages of animals in sediments (Som-

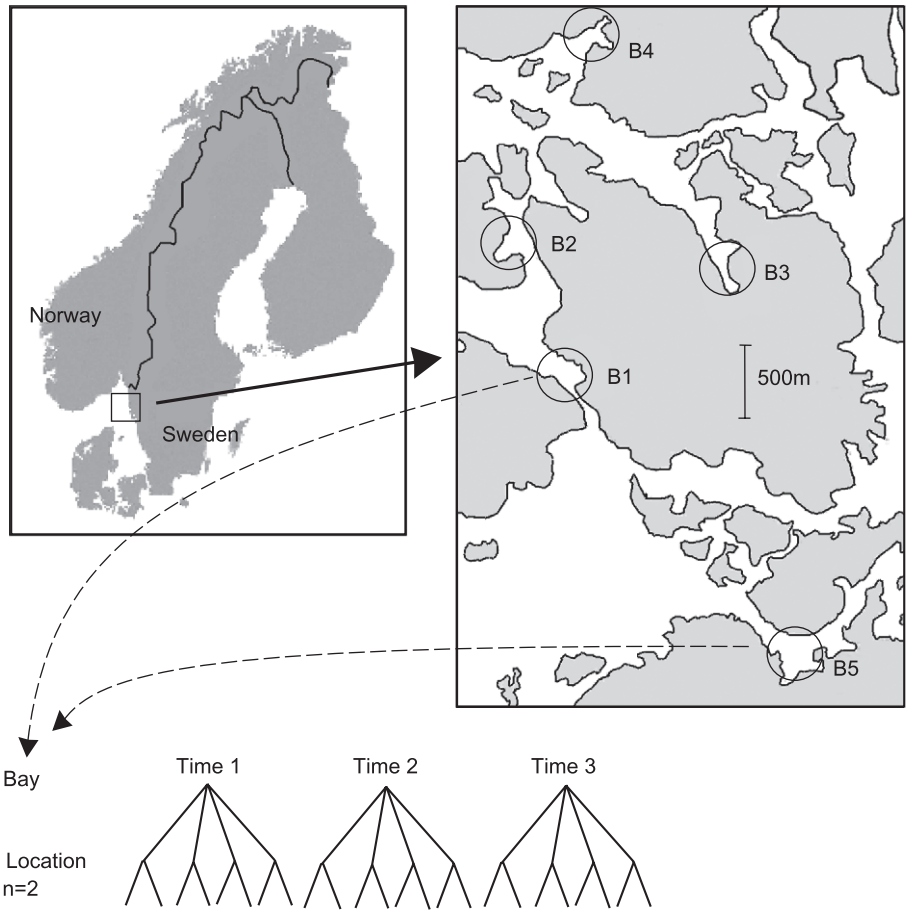

Fig. 1. Map of sampling area and sampling design. At three times, four locations within each of five bays (B_1–B_5) were sampled (n=2 replicates).

merfield and Clarke, 1995; Olsgard and Somerfield, 2000).

2.3. Sampling design and statistical analyses

The aim of this study was to test and estimate the importance of temporal and spatial components of variability in the structure and composition of soft-sediment infauna. Spatial variability was estimated at three hierarchical scales: among bays (10^3 m apart), among locations within bays (10^2 m apart), and among replicate samples (3–6 m=10^{0-1} m apart). In studies like these, any observations of significant variability at a particular scale may be caused by patchiness at scales smaller than that particular scale but larger than the next smaller scale. For example, observations of significant variability among location must be interpreted as patchiness at scales smaller than 10^2 m but larger than 10^{0-1} m. At each time of sampling, two sediment cores were taken at each location. In each bay, there were four locations that were revisited each time (Fig. 1). Sampling was done during 1 or 2 days at three times: November 2001, April 2002, and August 2002.

Hypotheses about variability in diversity and abundance of individual taxa among times and places were tested using three-factor analysis of variance (ANOVA). Times, bays and locations were selected at random to test the significance of their respective components of variability, including any interactive sources of variability. Tests of variability among bays are dependent on the elimination of components of

variability due to sites, or the interaction among times and bays (Underwood, 1997). Furthermore, spatial variability of diversity and abundances of individual taxa were tested and estimated at individual times using nested two-factor ANOVA. These analyses were done to achieve more detailed analyses of the nature of the frequently observed interactions between times and the different types of spatial variability. Cochran's test was used to test for heterogeneity of variances. If significant heterogeneity was detected, data were transformed by sqrt(X), sqrt(X+1), or $\log_{10}(X+1)$ (Underwood, 1997; Quinn and Keough, 2002).

The absolute and relative importance of spatial sources of variability was assessed by estimation of variance components according to procedures described in Underwood (1997). Negative estimates of variance components were set to zero and remaining components were recalculated according to procedures described by Fletcher and Underwood (2002).

Hypotheses about the spatial variability of whole assemblages were tested using nonparametric multivariate analyses of variance (NP-MANOVA) with restricted permutation of raw data (Anderson, 2001). Analyses were done separately for each time of sampling in order to evaluate any tendencies for interactive variability at the scale of bays, i.e. whether differences among bays varied among times, *a posteriori* tests of differences among bays were also done for each time. Spatial and temporal patterns of whole assemblages were illustrated using unconstrained ordination, using principal coordinate analysis (PCO). All multivariate analyses were done using $\log_{10}(X+1)$-transformed data and Bray–Curtis dissimilarity measures of dissimilarity (Bray and Curtis, 1957; Clarke, 1993).

3. Results

3.1. Univariate analyses

A total of 54 taxa were found during the three times of sampling. Twelve of these were found only in one of the 120 cores. The average number of taxa per core was nine (range 1–17). Components of variability were tested and estimated for: total number of individuals, total number of taxa, Shannon diversity $H'=-\sum P_i \log_{10}(P_i)$, where P_i=the proportion of the total sample belonging to the ith taxa, Margalef richness d=(number of taxa−1)/($\log_{10}$ total number of individuals), Pielou's evenness $J'=H'/(\log_{10}$ number of taxa), and the 15 most frequently observed taxa (present in ≥25% of the samples). Peet (1974) described and discussed the three diversity indices, which were calculated using the PRIMER-E software package (Clarke and Gorley, 2001). The most frequent taxa were oligochaetes, capitellid polychaetes, and hydrobid snails, which were found in more than 80% of the cores. These taxa were also most abundant, constituting 40%, 16%, and 14%, respectively, of the total abundance (Fig. 2).

Analyses of individual variables indicated that significant variability was observed for all sources of variability except among bays (Tables 1 and 2). The two interactions involving 'Time' in combination with 'Bay' and 'Location (B)' were most frequently observed significant (16 and 13 variables, respectively). Thus, changes in diversity and abundance of animals typically differ among bays and locations within bays. Out of 20 variables, all had at least one significant interaction. The variables describing collective properties of the assemblages: total abundance, number of taxa, H', d, and J' did not show similar interactive patterns. For the total abundance and all measures of diversity, the interactions between 'Time' and 'Bay' were significant. An example of what such an interaction can mean in practice is illustrated by the total number of taxa (Fig. 2). There was a consistent increase in the number of taxa from times 2 to 3 in bay 1, while in bay 3, there was a consistent decrease. Additionally for the total abundance, H', and d, there were significant interactions between times and locations within bays. One obvious example of such patterns is the change in richness in bay 5 at time 3. While richness decreases at three locations, there is an increase at the first location. Similarly, patterns of variability differ considerably among individual taxa (Table 2). The large number of significant interactions between bays and times indicates that changes in the abundance of infauna generally differ among bays. Interactions among locations and times, which were also frequent, indicate that the abundances of many taxa also change differently among locations within bays.

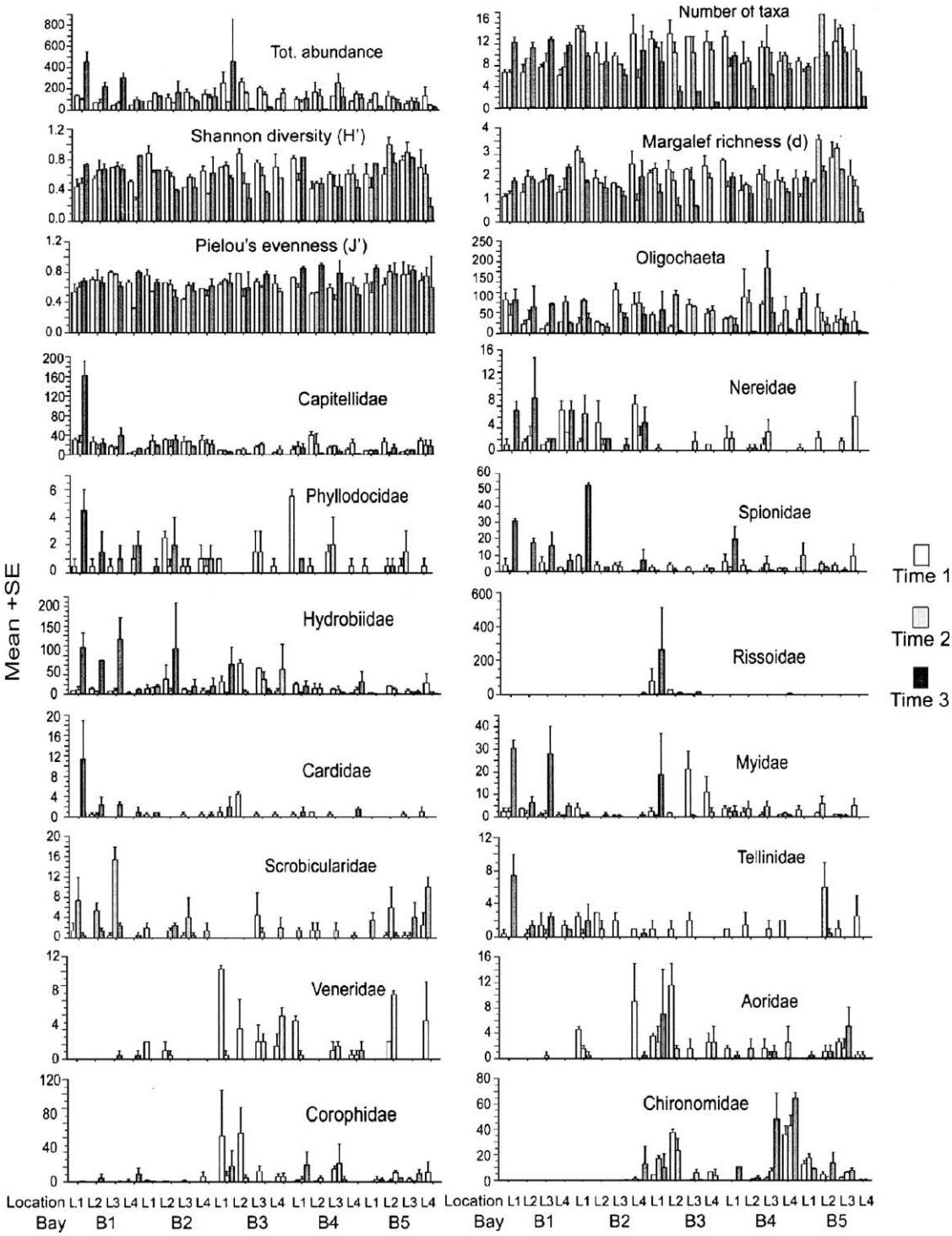

Fig. 2. Variation in mean (±1 S.E.) abundance of 15 taxa and values of five variables describing assemblage composition per location (n=2 replicate cores).

Table 1
Results from three-factor nested analysis of variance of number of taxa and total abundance

Source	*df*	Error term	Number of taxa			Total abundance		
			MS	*F*	*p*	MS	*F*	*p*
Time, *T*	2	*B***T*	75.41	1.27	0.33	0.79	1.02	0.40
Bay, *B*	4		6.03			0.31	0.40	0.80
*B***T*	8	*T***L*(*B*)	59.15	6.68	<0.01	0.78	3.45	0.01
Location (Bay), *L*(*B*)	15	*T***L*(*B*)	20.44	2.31	0.02	0.24	1.07	0.42
*T***L*(*B*)	30	Residual	8.86	1.29	0.20	0.23	3.88	<0.01
Residual	60		6.87			0.06		

Due to the large number of significant interactions with time, two-factor ANOVAs were done to gain further insight into the spatial variability at individual times. Analyses show that significance and relative contribution from different spatial scales differ among variables and times of sampling (Table 3, Fig. 3). Interestingly, only one variable, the chironomids, displays significant differences in abundance among locations within bays at all times of sampling. For none of the other types of animals and at none of the spatial scales was there consistent significant different variability at all times of sampling. There were, however, consistent nonsignificant variability among bays for *H*′ and *J*′ and scrobicularid bivales. The abundances of hydrobids, corophids, and cardid bivalves (only analysed at two times) were consistently nonvariable among locations. The analyses also show that significant variability among bays was most frequently observed at time 3, while variability among locations was frequent at times 2 and 3. Despite some exceptions, calculations of variance components suggest that components asso-

Table 2
Results from three-factor analysis of variance of abundance of 15 taxa and five variables characterising whole assemblages

Taxon	Source				
	Time	Bay	*T***B*	*L*(*B*)	*T***L*(*B*)
	p	*p*	*p*	*p*	*p*
Total abundance[b]	ns	ns	**	ns	**
Number of taxa	ns	nt	**	*	ns
H′	ns	nt	*	ns	**
d	ns	nt	**	*	*
J′[a]	ns	ns	ns	ns	**
Oligochaeta[b]	ns	nt	*	ns	*
Capitellidae[c]	ns	nt	**	*	*
Nereidae[a,d]	ns	nt	ns	ns	*
Phyllodocidae	ns	nt	*	ns	ns
Spionidae[d]	ns	nt	**	ns	**
Hydrobiidae[b]	ns	nt	**	ns	ns
Rissoidae[a,b]	ns	nt	**	**	ns
Cardidae[b]	ns	ns	**	ns	*
Myidae[a,b]	ns	ns	**	ns	ns
Scrobicularidae	*	ns	ns	ns	**
Tellinidae[d]	ns	ns	**	ns	**
Veneridae[a,d]	*	ns	ns	ns	**
Aoridae[a,b]	ns	nt	*	*	ns
Corophidae[d]	ns	nt	**	*	ns
Chironomidae[a,d]	ns	nt	**	**	**

Statistical significance from ANOVA analyses.
ns=not significant; nt=not tested; *p*=*p* value.
[a] Heterogenous variances.
[b] $\mathrm{Log}_{10}(X+1)$.
[c] $(X+1)^{0.5}$.
[d] $(X)^{0.5}$.
* $p<0.05$.
** $p<0.01$.

Table 3
Summary of results from two-factor analysis of variance of abundance of 15 taxa and five variables characterising whole assemblages

Taxon	Time 1		Time 2		Time 3	
	B	*L*(*B*)	*B*	*L*(*B*)	*B*	*L*(*B*)
	p	*p*	*p*	*p*	*p*	*p*
Total abundance	*	ns	ns	*	ns	*
Number of taxa	*	ns	ns	*	*	ns
H′	ns	ns	ns	*	ns	*
d	ns	ns	ns	*	*	*
J′	ns	ns	ns	*	ns	*
Oligochaeta	ns	ns	ns	*	*	*
Capitellidae	ns	*	ns	ns	*	*
Nereidae	ns	ns	ns	*	*	ns
Phyllodocidae	ns	*	ns	ns	*	ns
Spionidae	ns	ns	ns	ns	*	*
Hydrobiidae	ns	ns	ns	ns	*	ns
Rissoidae	nt	nt	nt	nt	*	*
Cardidae	ns	ns	nt	nt	*	ns
Myidae	ns	*	ns	ns	*	ns
Scrobicularidae	ns	ns	ns	*	ns	*
Tellinidae	ns	ns	nt	nt	*	ns
Veneridae	ns	ns	ns	*	nt	nt
Aoridae	ns	*	ns	ns	nt	nt
Corophidae	*	ns	ns	ns	ns	ns
Chironomidae	ns	*	*	*	*	*

Each time of sampling analysed separately. nt=not tested because of presence in less than 25% of sampled cores. Transformations and symbols as in Table 2.
* $p<0.05$.

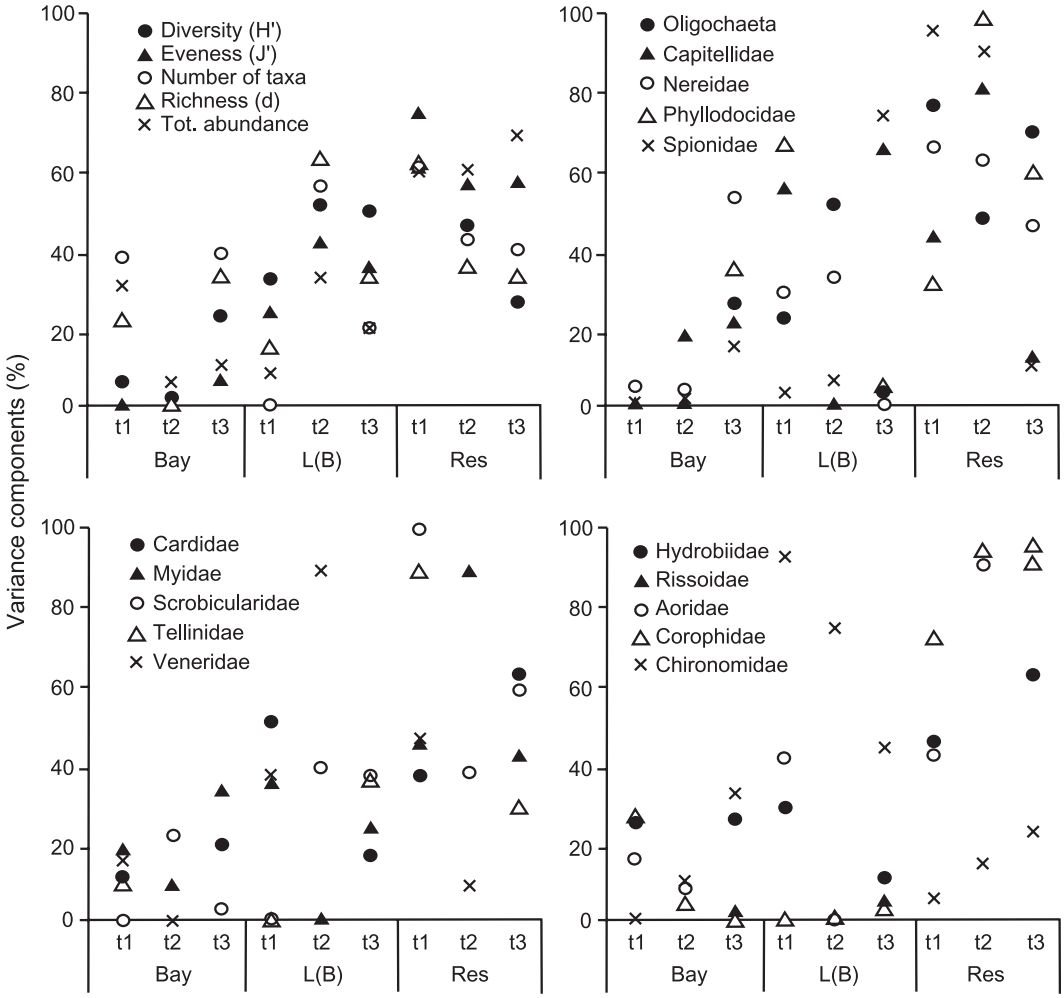

Fig. 3. Estimated variance components for two-factor analysis of variance performed for each time of sampling for dominant taxa and selected measures of diversity.

ciated with small spatial scales are generally more important than those associated with larger scales (Fig. 3).

3.2. Multivariate analyses

The results of the NP-MANOVA of assemblages at individual times consistently revealed significant spatial variability among bays and among locations within bays (Table 4). Estimates of the relative size of the different components of spatial variability suggested that variability among cores was consistently largest (42–64%). The relative importance of variability among locations and among bays, however, differed among times (Table 4).

Furthermore, analyses of the nature of spatial patterns at individual times using pairwise *a posteriori* tests showed that significant differences among bays in the structure of these assemblages were frequent (Table 4). Only for one of the pairs (bay 1 vs. bay 2) were differences among bays consistent at all times. Comparisons of one pair (bay 4 vs. bay 5) never revealed any significant differences in the structure of these assemblages. Conclusions about differences between the remaining eight combinations of bays differed among times of sampling. Thus,

Table 4
NP-MANOVA of spatial variability in benthic assemblages sampled in five bays at three times

Source	Time 1					Time 2				Time 3				Number of possible permutations
	df	MS	*p*	VC	%	MS	*p*	VC	%	MS	*p*	VC	%	
B	4	2568	*	184	19	2369	*	122	10	9545	*	954	39	2.5×10^9
L(*B*)	15	1097	*	282	28	1395	*	313	26	1917	*	448	18	1.0×10^{10}
Residual	20	532		532	53	770		770	64	1021		1021	42	

A posteriori tests among bays within times

	Time 1	Time 2	Time 3	
Bay 1 vs. bay 2	*	ns	*	35
Bay 1 vs. bay 3	*	*	*	35
Bay 1 vs. bay 4	ns	ns	*	35
Bay 1 vs. bay 5	ns	ns	*	35
Bay 2 vs. bay 3	*	ns	*	35
Bay 2 vs. bay 4	ns	ns	*	35
Bay 2 vs. bay 5	ns	ns	*	35
Bay 3 vs. bay 4	*	ns	*	35
Bay 3 vs. bay 5	*	ns	*	35
Bay 4 vs. bay 5	ns	ns	ns	35

Data transformed by $\log_{10}(X+1)$. A posteriori comparisons between pairs of bays were done using *t* and were not corrected for multiple comparisons.
Significant (at α=0.05) and ns=nonsignificant results in a posteriori tests are denoted by * and ns respectively.

although interactive sources of variation are not explicitly tested for the multivariate data (because analyses are done at individual times), the frequently observed inconsistencies among times suggest that the structure of these benthic assemblages changes differently among bays. This view is also supported in the graphical representations using principle coordinate analyses (Fig. 4). In consistency with the explicit tests, these graphs show that differences among bays are most pronounced at the third time of sampling and that differences among bays at the first time of sampling are largely caused by assemblages in bay 3, which are different from those in all other bays.

4. Discussion

The aim of this study was to estimate and test the significance of variability of infauna at multiple spatial scales and to evaluate how this spatial variability changed among times. The presence of significant spatial and temporal variability in all univariate and multivariate analyses suggests that these assemblages are strongly structured, possibly by both biological and physical processes. More specifically, it was concluded that interactive variability involving times of sampling and the two spatial scales were most frequently significantly variable. Thus, the abundance and composition of animals in bays situated a few kilometers apart and locations situated hundreds of meters apart generally changed differently among times approximately 4 months apart. Another general pattern was that the small-scale variability (i.e., among cores less than 3–6 m apart) was of considerable size compared to other sources of spatial variability. These results provide the first detailed accounts of interactive variability of infauna at multiple spatial scales in Swedish coastal waters and in soft sediments in general. More importantly, these results give insights into (1) the patterns and (2) processes affecting benthic infauna in the area, and (3) how to design more efficient programs for sampling these assemblages in future studies.

The ecological meaning of the prevalence of interactive variability is that the abundance and composition of benthic assemblages change differently among times of sampling in different bays and

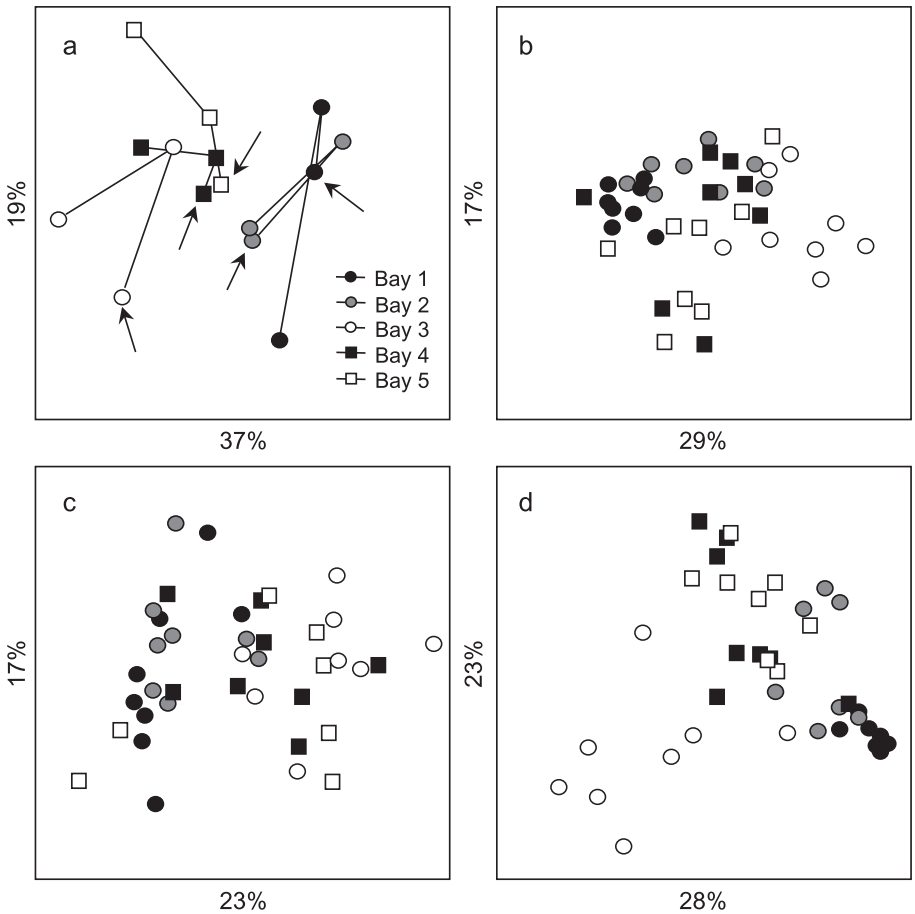

Fig. 4. Principal coordinate analysis of (a) centroids for each bay at each time; (b, c, and d) samples at times 1, 2, and 3, respectively. Data transformed to $\log_{10}(X+1)$. Arrows in (a) indicate position of initial time of sampling and percentages are proportion of variance explained by axes 1 and 2.

locations within bays. One consequence of this is that the scale-dependent variability (i.e., patchiness) appears to change strongly for individual taxa among times of sampling. Thus, there are not only large fluctuations in absolute numbers of animals, but there are also changes in the intensity and relative importance of different spatial scales among times. This is particularly evident from the lack of consistent significant variability among bays or locations at different times of sampling for any of the univariate measures. This is surprising considering the short interval among times of sampling (4–5 months), in relation to the generation time of many of the animals (e.g., Connell and Sousa, 1983). Nevertheless, it is consistent with some other studies on benthic assemblages at medium to short time scales (e.g., Morrisey et al., 1992b; Olabarria and Chapman, 2001; Lindegarth and Underwood, 2002). Interactive variability was not explicitly evaluated for multivariate patterns. Comparisons among locations, however, showed that assemblages generally differed between locations at some times but not at others. Despite this apparent prevalence of interactive variability also in the multivariate analyses, differences between one pair and lack of differences between another were consistent among times. These findings are partly consistent with what was concluded by Turner et al. (1995), who concluded that multivariate patterns were more constant than those of individual taxa. The relative constancy of multivariate patterns is some-

what surprising, but it may be explained by the fact that these reflect aspects of composition as well as abundance (e.g., Connell and Sousa, 1983; Micheli et al., 1999). Thus, if there is a persistent, but variable in abundance, presence of some species at some locations but absence at others, this may have strong effects on multivariate patterns.

A first assessment of the causes of these patterns of interactive variability among bays and times of sampling suggests that the processes involved differ between the two periods (November–April vs. April–August). For example, the polychaetes, gastropods, and some bivalves generally decreased in abundance during the first period (Fig. 5). During the following period, the abundances of these taxa increased at variable rates in some bays and decreased in others. This suggests that processes causing mortality (or emigration) during the first period were spatially more consistent among bays than those causing increased abundances during the second period. This result is consistent with the view that factors associated with cold winter temperatures profoundly and generally regulate assemblages of animals in shallow boreal areas (e.g., Möller and Rosenberg, 1983). The intensity of processes causing growth in populations, either by local reproduction or immigration into these habitats during the second period, was more variable at the scale of bays. Thus, as in other areas, it appears that processes associated with the supply of larvae, settlement, and survival of recruits are strongly spatially variable (Möller and Rosenberg, 1983; Gaines and Bertness, 1993; Lindegarth et al., 1995; Bouma et al., 2001). These dynamic patterns were, however, not representative for all types of animals. For example, the most abundant taxon, the oligochaetes, increased fairly consistently in all bays during the first period, but showed a strongly variable decrease during the second (Fig. 5). Thus, the dynamics of oligochaetes appears to be driven by processes varying at other spatial scales than what was apparent for other groups of animals. Furthermore, the overall dynamics of these assemblages measured as

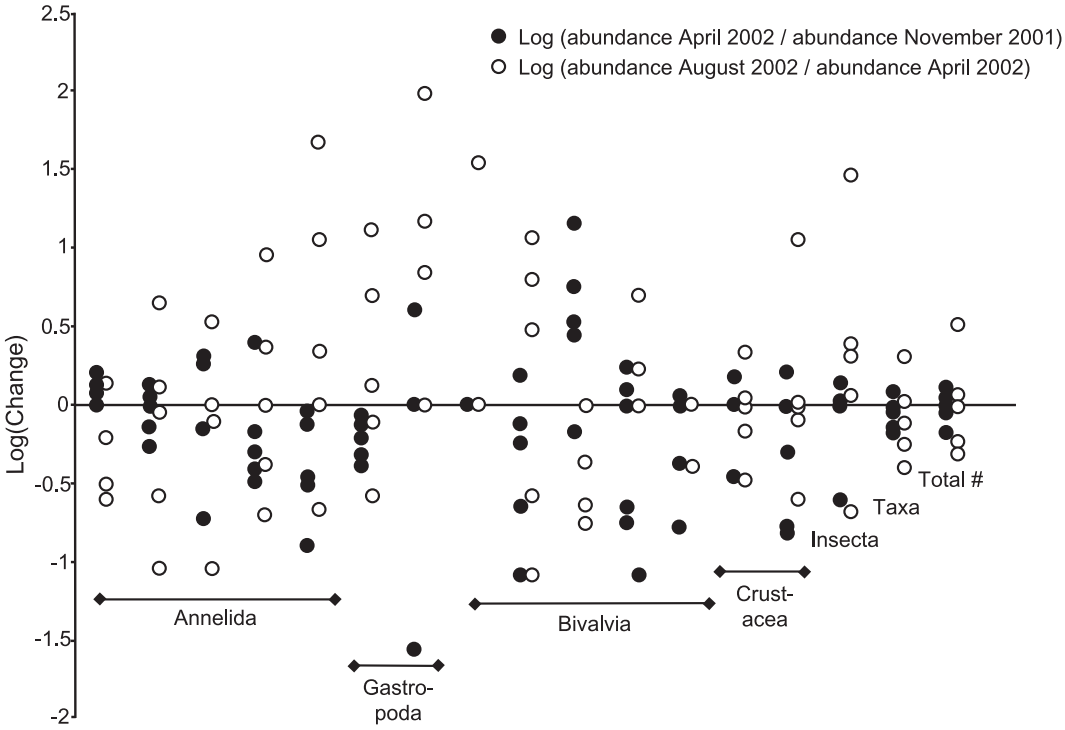

Fig. 5. Change in abundance in each bay among times of sampling. Taxa are from left to right: annelids (oligochaetes, capitellids, nereids, phyllodocids, and spionids), gastropods (*Hydrobia* sp. and *Rissoa* sp.), bivalves (myidae, scrobicularidae, tellinidae, and veneridae), crustaceans (aoridae and corophidae), and insects (chironomidae).

changes in number of taxa and total number of individuals per core suggests that changes were more consistent among bays during the first period than during the second. During April to August, some bays decreased, while others increased more extremely than any of the bays did from November to April.

Another general characteristic of the patterns of these assemblages was the prevalence of large variability among cores within individual locations and times. This variability may reflect random processes acting on individual animals, but it may also include processes causing spatial structure at scales smaller than those tested in this study (i.e., locations within bays). One way to assess the relative importance of structuring vs. completely random processes at the scale of individual cores is to evaluate the relationship between the mean abundances and the coefficient of variation. If the variability among cores was not caused by any structuring process, the relationship between those two parameters is expected to be similar to that of a Poisson process (Underwood, 1997; Quinn and Keough, 2002). Without doing any formal tests, it appears that the coefficient of variation for some species (e.g., capitellid and nereid polychaetes) is distributed in a way which is what would be expected from a purely random process at most locations and times (Fig. 6). Other taxa, such as the oligochaetes and the hydrobid snails, are often substantially more variable than what might be expected if individual animals were randomly distributed. Similar to what was found in other studies, this clearly suggests that the abundances of many taxa are spatially structured at scales smaller than 5 m, despite apparent environmental homogeneity (e.g., Eckman, 1979; Bergström et al., 2002). At these scales, the importance of biological interactions in structuring assemblages and populations in soft sediments has been repeatedly demonstrated through manipulative experiments. These interactions include competition (Wilson, 1991), predation by epifauna and infauna (André and Rosenberg, 1991; Olafsson et al., 1994), active responses to settlement cues (But-

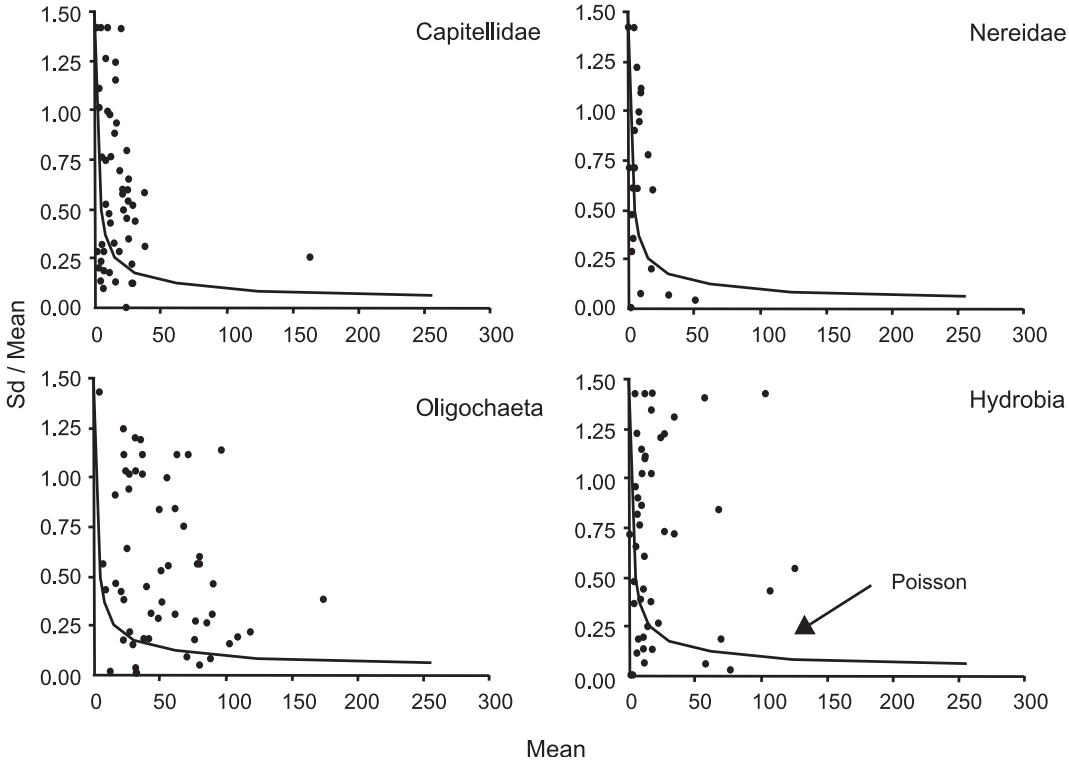

Fig. 6. Observed and predicted (Poisson) relationships between mean abundance and coefficient of variation at individual locations and times.

man, 1987), and responses to patchy food resources (e.g., Nilsson et al., 1997; Kelaher and Levinton, 2003). It is, however, also possible that purely physical mechanisms, such as passive deposition of larvae, small-scale hydrodynamic disturbance, and local topography contribute to patterns at these small spatial scales (Eckman, 1979; Jonsson et al., 1991). Similarly, the importance of biologically driven processes for the formation of larger-scale patterns previously discussed cannot be ruled out.

In order to estimate the magnitude and test the significance of variability at multiple spatial scales, it is fundamental and necessary requirement for sampling to be designed in a way that allows unconfounded estimation of the variance components of interest. For any given level of effort at each time of sampling (here approximately 45 days were used for sampling, sorting, and identification), however, there exists a large number of possible ways to allocate efforts among the different spatial scales. These different possible solutions are going to differ with respect to statistical power and precision, and any selected sampling design represents a compromise among scales with respect to power, accuracy, and precision.

The large frequency of significant interactions between times of sampling and the different spatial scales is a clear indication that the selected experimental design was precise enough to detect differences in spatial variability among times (Tables 1–4). Furthermore, the precision of estimated means can be calculated from estimates of variance components (Table 5). The typical estimated precision of overall means for individual taxa in the investigated region is in the range 0.1–0.45 (S.E./mean), while for individual bays and locations, the corresponding values are 0.3–0.9 and 0.5–1.8, respectively. These results can be contrasted to a sampling design, which is optimised with respect to the precision of the overall mean estimate (see Underwood, 1981).

It is evident that given the observed costs associated with different aspects of sampling and patterns of spatial variability, the optimal way to estimate the overall mean involves sampling of only one replicate core at each location within each bay. This result is consistent for all taxa (Table 5). The optimal allocation of samples among locations and bays, however, differs strongly among taxa. In general, the results suggest that a relatively large

Table 5
Results of cost–benefit analyses[a] for individual taxa

Taxon	Optimal design			Optimal			Observed		
				S.E./mean			S.E./mean		
	n	*b*	*a*	Region	Bay	Location	Region	Bay	Location
Oligochaeta	1	14	3	0.11	0.19	0.67	0.12	0.27	0.47
Capitellidae	1	2	19	0.15	0.61	0.68	0.18	0.36	0.48
Spionidae	1	14	3	0.17	0.28	0.86	0.19	0.43	0.61
Nereidae	1	1	34	0.27	1.45	1.36	0.39	0.54	0.96
Phyllodocidae	1	14	3	0.24	0.42	1.47	0.26	0.58	1.04
Myidae	1	14	3	0.24	0.41	1.54	0.25	0.55	1.09
Scrobicularidae	1	2	19	0.25	0.98	1.35	0.30	0.50	0.96
Tellinidae	1	14	3	0.22	0.38	1.41	0.22	0.50	0.99
Veneridae	1	1	34	0.27	1.41	1.41	0.38	0.50	1.00
Cardidae	1	14	3	0.40	0.68	2.55	0.41	0.91	1.80
Corophidae	1	2	19	0.42	1.78	2.42	0.46	0.93	1.71
Aoridae	1	2	19	0.34	1.32	1.86	0.42	0.66	1.31
Hydrobiidae	1	14	3	0.22	0.38	1.40	0.22	0.50	0.99
Rissoidae	1	4	10	1.20	3.63	6.81	1.32	2.72	4.82
Chironomidae	1	2	19	0.29	1.14	0.83	0.41	0.75	0.59

Precision of estimated means in the region, bays, and locations is presented as S.E./overall mean for each taxon. The total time (cost) available for each time of sampling was 45 days (360 h). Costs for additional replicates, locations, and bays were 8, 0.33, and 2 h, respectively. *n*=number of replicates; *b*=number of locations; and *a*=number of bays.

[a] Optimisations were done with the restriction that the number of bays, $a \geq 2$.

number of locations should be sampled to achieve precise estimates of taxa, which vary strongly among locations. In contrast, sampling in a large number of bays is optimal for taxa that are variable among bays. Thus, the selected sampling design was more intensive at individual locations but less intensive with respect to the numbers of locations or bays compared to the one that is optimal with respect to precision of the overall mean. The result of these differences in sampling design in terms of actual precision varies among spatial scales (Table 5). Not surprisingly, the optimal design is consistently more precise for estimates of the overall mean. Interestingly, however, the selected design was consistently more precise at individual locations and, for approximately 50% of the taxa, more precise in individual bays. Whether these expected levels of precision are sufficient obviously depends on the context of the study. General recommendations about desired levels of precision are difficult to make, but a ratio of 0.3 (S.E./mean) is sometimes used (Andrew and Mapstone, 1987). The results presented here suggest that this is achieved for many taxa at the scale of the entire region for the optimal as well as the selected design (Table 5).

These results provide a valuable background for further sampling of the diversity of benthic assemblages in Swedish coastal areas. Such studies will be done to test the generality of spatial patterns and to develop statistical models for prediction of abundance and composition of these assemblages. Information about variability can be used to evaluate what the likely success of such models is at different spatial and temporal scales.

Acknowledgements

We would like to thank Lars Gamfeldt for assistance in the field and in the laboratory. The manuscript was greatly improved by comments from two anonymous reviewers. This work was funded by the Swedish Environmental Protection Agency (MARBIPP), FORMAS (contract 21.0/2001-4015), and the European Regional Development Fund (ERDF). Additional funding was supplied by the Colliander foundation. **[AU]**

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